

Equation Selection Analysis for Cardiac Drug Response Modeling

Cardiac RODEO Project

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Abstract

This report presents a comprehensive analysis of 12 candidate equations for modeling cardiac drug response data. Each equation was fit to both Contractility and O_2 consumption measurements across 25 drugs. The equations are evaluated based on their coefficient of determination (R^2), and parameter interpretability.

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1 Executive Summary

1.1 Top Performing Equations

Table 1: Top 5 Equations by Mean R^2

Rank	Equation	Contractility	O ₂
1	Dual Exponential (Eq1)	0.4971	0.5656
2	Modified Hill (Hormesis) (Eq4)	0.4509	0.5440
3	Hormesis V0 (Legacy) (Eq12)	0.4265	0.4758
4	Biphasic Response (Eq7)	0.3993	0.4836
5	PKPD Elimination (Eq11)	0.1809	0.5026

Table 2: Top 3 Equations by O₂ Mean R^2

Rank	Equation	O ₂ Mean	O ₂ Std
1	Dual Exponential (Eq1)	0.5656	0.2297
2	Modified Hill (Hormesis) (Eq4)	0.5440	0.2273
3	PKPD Elimination (Eq11)	0.5026	0.2538

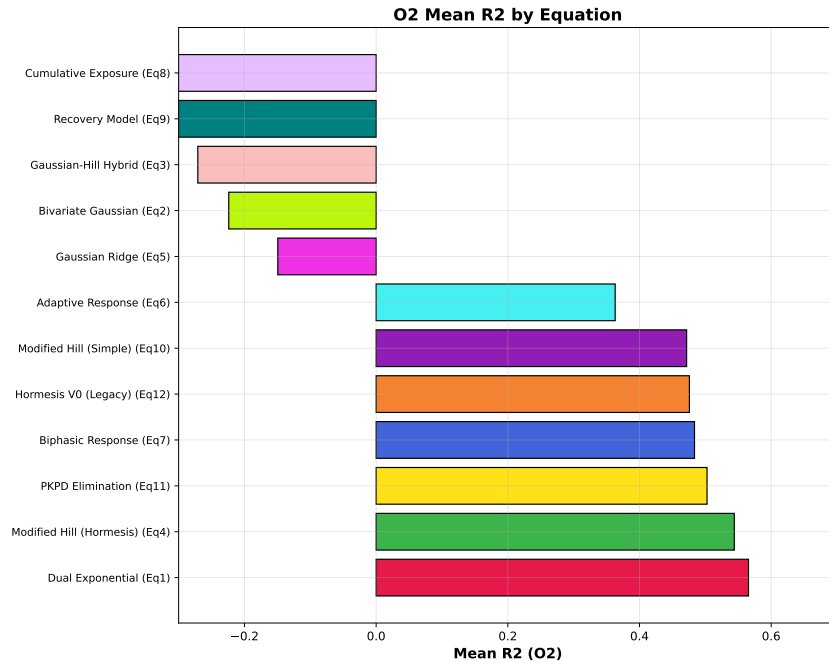


Figure 1: Mean R^2 values for O₂ by equation.

2 Candidate Equations

2.1 Dual Exponential (Eq1)

Description: Separate exponential terms for beneficial and toxic effects.

Mathematical Form:

$$R(C, t) = R_0 + A_{\text{benefit}} \left(1 - e^{-k_b C^{n_b}}\right) (1 - e^{-t/\tau_b})^{m_b} - A_{\text{tox}} \left(1 - e^{-k_t C^{n_t}}\right) (1 - e^{-t/\tau_t})^{m_t} \quad (1)$$

Parameters: $R_0, A_{\text{benefit}}, A_{\text{tox}}, k_b, k_t, \tau_b, \tau_t, n_b, n_t, m_b, m_t$

2.2 Bivariate Gaussian (Eq2)

Description: Sum of two bivariate Gaussian surfaces for localized effects.

Mathematical Form:

$$R(C, t) = R_0 + A_1 G_1(C, t) + A_2 G_2(C, t) \quad (2)$$

$$\text{where } G_i(C, t) = \exp \left(-\frac{1}{2(1-\rho_i^2)} \left[\frac{(C-\mu_{C,i})^2}{\sigma_{C,i}^2} + \frac{(t-\mu_{t,i})^2}{\sigma_{t,i}^2} - \frac{2\rho_i(C-\mu_{C,i})(t-\mu_{t,i})}{\sigma_{C,i}\sigma_{t,i}} \right] \right)$$

Parameters: $R_0, A_1, A_2, \mu_{C,1}, \mu_{T,1}, \sigma_{C,1}, \sigma_{T,1}, \rho_1, \mu_{C,2}, \mu_{T,2}, \sigma_{C,2}, \sigma_{T,2}, \rho_2$

2.3 Gaussian-Hill Hybrid (Eq3)

Description: Combines Hill concentration dependence with Gaussian time profile.

Mathematical Form:

$$R(C, t) = R_0 + E_{\text{max}} H(C) G(t) - E_{\text{tox}} T(C) f(t) \quad (3)$$

where:

- $H(C) = \exp \left(-\frac{(C-\mu_C)^2}{2\sigma_C^2} \right)$ (Gaussian concentration profile)
- $G(t) = \frac{(t/\tau)^m}{1+(t/\tau)^m}$ (Hill-type time dependence)
- $T(C) = \frac{C^n}{TC_{50}^n + C^n}$ (toxic Hill function)
- $f(t) = 1 - e^{-t/\tau_{\text{tox}}}$ (time onset of toxicity)

Parameters: $R_0, E_{\text{max}}, \mu_C, \sigma_C, \tau, m, E_{\text{tox}}, n, TC_{50}, \tau_{\text{tox}}$

2.4 Modified Hill (Hormesis) (Eq4)

Description: Modified Hill equation with hormesis (low-dose benefit, high-dose toxicity).

Mathematical Form:

$$R(C, t) = R_0 + E_{\text{benefit}} H_b(C) f_b(t) - E_{\text{tox}} H_t(C) f_t(t) \quad (4)$$

where:

- $H_b(C) = \frac{C^{n_b}}{EC_{50,b}^{n_b} + C^{n_b}}$ (benefit Hill function)
- $H_t(C) = \frac{C^{n_t}}{TC_{50}^{n_t} + C^{n_t}}$ (toxicity Hill function)
- $f_b(t) = 1 - e^{-t/\tau_b}, f_t(t) = 1 - e^{-t/\tau_t}$ (time onset functions)

Parameters: $R_0, E_{\text{benefit}}, E_{\text{tox}}, EC_{50,b}, TC_{50}, n_b, n_t, \tau_b, \tau_t$

2.5 Gaussian Ridge (Eq5)

Description: Simple 2D Gaussian ridge centered at optimal concentration-time.

Mathematical Form:

$$R(C, t) = R_0 + A \exp \left(-\frac{(C - \mu_C)^2}{2\sigma_C^2} - \frac{(t - \mu_t)^2}{2\sigma_t^2} \right) \quad (5)$$

Parameters: $R_0, A, B, \mu_C, \sigma_C, \mu_{tox}, \sigma_{tox}, \kappa, \tau, m, \lambda$

2.6 Adaptive Response (Eq6)

Description: Models initial effect with adaptive tolerance over time.

Mathematical Form:

$$R(C, t) = R_0 + E_{\max} \frac{C^n}{EC_{50}^n + C^n} \cdot e^{-t/\tau_{\text{adapt}}} (1 - e^{-t/\tau_{\text{onset}}}) \quad (6)$$

Parameters: $R_0, E_{\max}, EC_{50}, n, \tau_{\text{onset}}, \tau_{\text{adapt}}$

2.7 Biphasic Response (Eq7)

Description: Biphasic model with low-dose stimulation and high-dose inhibition.

Mathematical Form:

$$R(C, t) = R_0 + E_{\text{stim}} H_1(C) f_1(t) - E_{\text{inhib}} H_2(C) f_2(t) \quad (7)$$

where:

- $H_1(C) = \frac{C^{n_1}}{EC_{50, \text{stim}}^{n_1} + C^{n_1}}$ (stimulation Hill function)
- $H_2(C) = \frac{C^{n_2}}{IC_{50}^{n_2} + C^{n_2}}$ (inhibition Hill function)
- $f_1(t) = 1 - e^{-t/\tau_{\text{stim}}}$ (stimulation time onset)
- $f_2(t) = 1 - e^{-t/\tau_{\text{inhib}}}$ (inhibition time onset)

Parameters: $R_0, E_{\text{stim}}, E_{\text{inhib}}, EC_{50, \text{stim}}, IC_{50}, n_1, n_2, \tau_{\text{stim}}, \tau_{\text{inhib}}$

2.8 Cumulative Exposure (Eq8)

Description: AUC-dependent toxicity based on cumulative drug exposure.

Mathematical Form:

$$R(C_0, t) = R_0 - E_{\text{tox}} \left(1 - e^{-\alpha \cdot \text{AUC}(t)/TC_{50}} \right) \quad (8)$$

where $\text{AUC}(t) = \frac{C_0}{k_{\text{elim}}} (1 - e^{-k_{\text{elim}} t})$ is the area under the concentration-time curve.

Parameters: $R_0, E_{\text{tox}}, \alpha, TC_{50}, k_{\text{elim}}$

2.9 Recovery Model (Eq9)

Description: Reversible damage with exponential recovery kinetics.

Mathematical Form:

$$R(C, t) = R_0 - E_{\text{damage}} (1 - e^{-k_{\text{damage}} C t}) \cdot e^{-k_{\text{recovery}} t} \quad (9)$$

Parameters: $R_0, E_{\text{damage}}, k_{\text{damage}}, k_{\text{recovery}}$

2.10 Modified Hill (Simple) (Eq10)

Description: Simplified Hill model with combined concentration-time effect.

Mathematical Form:

$$R(C, t) = R_0 + E_{\max} \left(1 - e^{-\kappa C^n (t/\tau)^m} \right) \quad (10)$$

Parameters: $R_0, E_{\max}, \kappa, \tau, n, m$

2.11 PKPD Elimination (Eq11)

Description: PK-PD model with first-order drug elimination kinetics.

Mathematical Form:

$$R(C_0, t) = R_0 + E_{\max} \left(1 - e^{-\kappa (C_0 e^{-k_{\text{elim}} t})^n (t/\tau)^m} \right) \quad (11)$$

where $C(t) = C_0 e^{-k_{\text{elim}} t}$ is the drug concentration at time t following first-order elimination.

Parameters: $R_0, E_{\max}, \kappa, n, m, \tau, k_{\text{elim}}$

2.12 Hormesis V0 (Legacy) (Eq12)

Description: Legacy hormesis model (similar to modified Hill hormesis).

Mathematical Form:

$$R(C, t) = R_0 + E_{\text{benefit}} H_b(C) f_b(t) - E_{\text{tox}} H_t(C) f_t(t) \quad (12)$$

where:

- $H_b(C) = \frac{(C/EC_{50,b})^{n_b}}{1 + (C/EC_{50,b})^{n_b}}$ (benefit Hill function)
- $H_t(C) = \frac{(C/TC_{50})^{n_t}}{1 + (C/TC_{50})^{n_t}}$ (toxicity Hill function)
- $f_b(t) = 1 - e^{-t/\tau_b}, f_t(t) = 1 - e^{-t/\tau_t}$ (time onset functions)

Parameters: $R_0, E_{\text{benefit}}, E_{\text{tox}}, EC_{50,b}, TC_{50}, n_b, n_t, \tau_b, \tau_t$

3 Complete Results



Figure 2: Heatmap of mean R^2 values. Green = good fit, Red = poor fit.

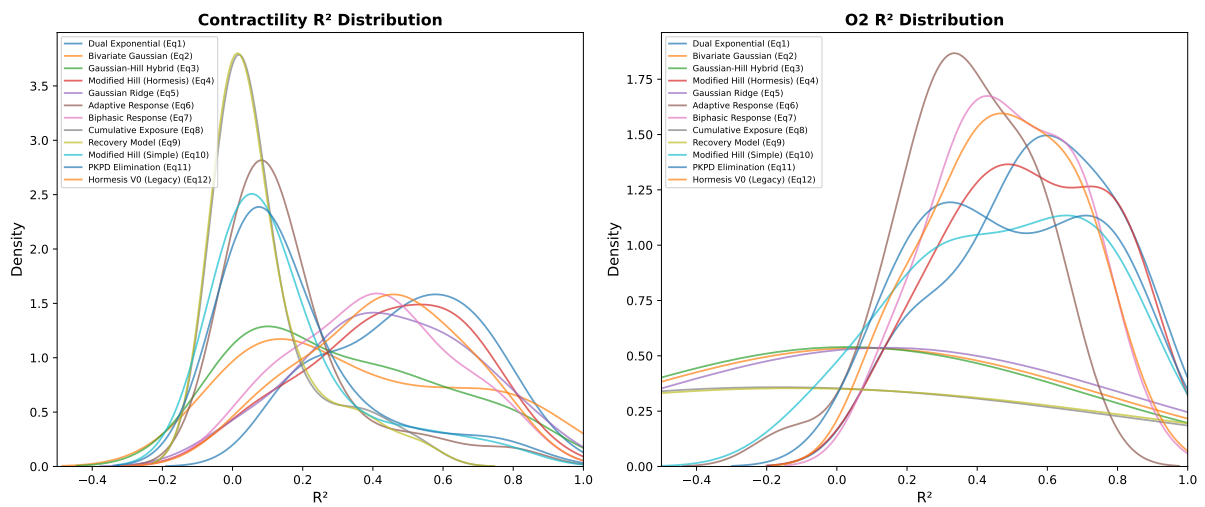


Figure 3: Distribution of R^2 values across drugs.

3.1 Full Statistics Table

Table 3: Complete R^2 Results by Equation

Equation	Cont. Mean	Cont. Std	O ₂ Mean	O ₂ Std
Dual Exponential (Eq1)	0.4971	0.2120	0.5656	0.2297
Modified Hill (Hormesis) (Eq4)	0.4509	0.2264	0.5440	0.2273
Hormesis V0 (Legacy) (Eq12)	0.4265	0.2184	0.4758	0.2017
Biphasic Response (Eq7)	0.3993	0.2207	0.4836	0.1918
PKPD Elimination (Eq11)	0.1809	0.2155	0.5026	0.2538
Modified Hill (Simple) (Eq10)	0.1541	0.2085	0.4716	0.2808
Adaptive Response (Eq6)	0.1638	0.1922	0.3631	0.1916
Gaussian Ridge (Eq5)	0.4462	0.2394	-0.1494	1.2168
Bivariate Gaussian (Eq2)	0.3625	0.3081	-0.2239	1.2270
Gaussian-Hill Hybrid (Eq3)	0.3188	0.2822	-0.2709	1.2355
Recovery Model (Eq9)	0.0854	0.1426	-0.6276	1.9000
Cumulative Exposure (Eq8)	0.0897	0.1450	-0.6666	1.8863

4 Conclusions

Based on the analysis of 12 equations:

1. The **Dual Exponential (Eq1)** equation achieved the highest overall performance with mean R^2 values of 0.4971 (Contractility) and 0.5656 (O_2).
2. Equations with separate benefit and toxicity components generally performed well.
3. The PK-PD elimination model provides interpretable pharmacokinetic parameters.

4.1 Recommendations

For cardiotoxicity prediction, we recommend:

- **Dual Exponential (Eq1)**: Mean $R^2 = 0.5313$
- **Modified Hill (Hormesis) (Eq4)**: Mean $R^2 = 0.4975$
- **Hormesis V0 (Legacy) (Eq12)**: Mean $R^2 = 0.4512$